Code No: 133AJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
DIGITAL LOGIC DESIGN
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Express the following numbers in decimal : $(10110.0101)_2$, $(16.5)_{16,}$ $(26.24)_8$.
b) Simplify the following Boolean expressions to a minimum number of literals
(i) ABC+A'B+ABC' (ii) xy + x(wz+wz') [7+8]

2.a) What are universal logic gates? Explain how do you realize an XOR gate and an Half adder using NAND gates only with necessary diagrams.
b) State and prove the D' Morgan theorems. Simplify the Boolean expressions to minimum number of literals (i) $x\,y\,z + x'\,y + x\,y\,z'$ (ii) $(x+y)\,(x+y')$. [7+8]

3.a) Write the simplification rules of Karnaugh's maps and simplify the Boolean function using 4 variable maps $F(A,B,C,D) = \sum(0,1,2,4,5,7,11,15)$
b) Simplify the Boolean function $F(w, x, y, z) = \sum (1,3,7,11.15)$ having don't care conditions $d(w,x,y,z) = \sum (0,2,5)$ and implement the output using NAND and NOR logic gates. [7+8]

4.a) Explain the design procedure for BCD to excess-3 code converter and draw the truth tables, maps and logic diagram.
b) Draw the circuit diagram of 4-bit adder/subtractor and explain its working by taking suitable examples. [8+7]

5.a) Explain the design procedure for multiplexers and de-multiplexers and draw the logic diagram of a 4-to-1 line multiplexer with logic gates.
b) Explain the functions of decoders. Construct 3-to-8 line decoder with logic gates with enable input. [8+7]

6.a) Describe the design analysis of clocked sequential circuits with help of neat diagram and explain its working with state equations, state tables and state diagrams.
b) Design a counter with the following repeated binary sequence: 0, 1, 2, 3, 4, 5, 6 use JK flip-flops. [8+7]

7.a) Given 32 × 8 ROM with enable input, Show the external connections necessary to construct a 128 × 8 ROM with 4 chips and a decoder.
b) Design a combinational circuit using a ROM that accept a 3-bit number and generates an output binary equal to the square of the input number. [8+7]

8.a) Explain the working of PAL and sequential programmable devices with help of neat diagrams and mention the advantages of programmable logic devices.
b) Explain with a schematic and implement the following two Boolean functions with a PLA: $F_1(A, B, C) = \sum (0, 1, 2, 4)$ and $F_2(A, B, C) = \sum(0, 5, 6, 7)$. [8+7]

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